Contact

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Top Skills

Machine Learning

Python (Programming Language)

Statistical Modeling

Honors-Awards

National Scholarship

Outstanding Graduate

Jiangsu Province Outstanding Student

Publications

Crystal growth, transport, and magnetic properties of antiferromagnetic semimetal $Ni_3In_2Se_2$

First-principles studies on the crystal structure and electronic properties of two-dimensional RuO_2 and OsO_2

Mode-selective anharmonicity induced by lone-pair electrons in layered oxyselenides

Patents

Analysis System and Method Based on First-Principles and Quantum Oscillation Testing

Wentworth LIU

Data Scientist | Applied ML, Statistical Modelling & Decision Science | Li Auto · ByteDance | Open to London Opportunities

London Area, United Kingdom

Summary

I am a Data Scientist focused on applying machine learning, statistical modelling, and decision science to complex operational and product problems.At Li Auto's Battery Intelligence department, I work with million-scale fleet telemetry to build causal analytics, explainable risk-ranking, reusable feature systems, and NLP workflows that support product and after-sales decisions. Before Li Auto, I worked in data analytics at ByteDance's Douyin Market Strategy team, where I built multi-source measurement and monitoring systems for content and campaign decisions.My academic background in computational materials science provides a rigorous foundation in mathematical modelling, statistical inference, scientific computing, and research validation. Across industrial and internet settings, I focus on turning ambiguous data problems into analytical systems and decision tools that teams can use repeatedly.Currently open to Data Scientist, Applied Scientist, Decision Science, and quantitative analytics opportunities in London.

Experience

Li Auto

Data Scientist

July 2024 - Present (2 years 4 months)

Beijing, China

Causal Inference & Decision Support: Used quasi-causal analysis on million-scale fleet telemetry to identify high-risk SOH-degradation cohorts and actionable drivers. Productised intervention analysis across multiple vehicle models, supporting targeted strategies projected to slow degradation and reduce warranty exposure by tens of millions of RMB for a single model.

Risk Modelling & Prioritisation: Recast thermal-runaway detection under extreme class imbalance as an explainable ranking task. Built a cross-platform ML pipeline with vehicle-level risk scores and feature-level drivers across multiple battery platforms, achieving 93% Recall@2% on retrospectively

validated high-risk cases while limiting inspection to the highest-risk 2% of vehicles.

Feature Engineering & Data Productisation: Built a configurable feature system using custom event definitions, multi-operator aggregations, and cross-platform normalisation. Deployed reusable workflows to an internal ML platform across multiple battery platforms, improving feature-iteration efficiency by ~85%.

NLP Classification & Workflow Automation: Built a hybrid classifier for battery-quality work orders across 30+ fault categories using rule-based matching, TF-IDF, and Random Forest. Achieved Weighted F1 > 0.90 and introduced confidence-based routing to focus review effort on ambiguous cases.

ByteDance
Data Analytics
May 2023 - September 2023 (5 months)
Beijing, China

Business Metrics & Monitoring: Built a multi-source IP popularity measurement system by integrating consumption, dissemination, and search signals across multiple data platforms. Added automated pipeline monitoring to address upstream data instability and support recurring content-investment and promotional decisions.

Funnel Analysis & Growth Strategy: Redesigned performance metrics and built a multi-dimensional content taxonomy for Qishui Music. Diagnosed new-user conversion decline through funnel segmentation and KOL audience-saturation modelling, informing content selection and marketing-resource allocation.

Dongwu Securities Co.
Equity Research Analyst
December 2021 - May 2022 (6 months)
Nanjing, Jiangsu, China

Produced an equity-research report on Harmonic Drive Systems, combining company fundamentals, competitive analysis, and market dynamics with quantitative stock-price modelling to deliver short-, medium-, and long-term investment recommendations.

Education

Nanjing University

Master of Science, Computational Materials Science · (September 2021 - June 2024)

Nanjing University

Bachelor of Science, Materials Science & Engineering · (September 2017 - July 2021)